A

MINI PROJECT REPORT

ON

"Temperature and Humidity Monitoring System Using Arduino Nano And DHT11"

In Partial Fulfillment of the Requirement for the Bachelor of Technology

Second Year, Third Semester in Electronics & Telecommunication Engineering

Submitted

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Under the Guidance of

Prof. A.R.DUDHE

Department of Electronics & Telecommunication Engineering

JAGADAMBHA COLLEGE OF ENGINEERING & TECHNOLOGY,

YAVATMAL – 445001

2025-2026

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CERTIFICATE

This is to certify that the mini project report entitled ***"Temperature and Humidity Monitoring System Using Arduino Nano And DHT11"****is a bonafide work done under our supervision and is submitted to Electronics & Telecommunication Engineering Engineering Department, Jagadambha College of Engineering and Technology, Yavatmal, An Autonomous Institution affiliated to Sant Gadge Baba Amravati University, Amravati in partial fulfillment of the requirement for the Bachelor of Technology - Second Year, Third Semester in Electronics & Telecommunication Engineering Engineering*

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ABSTRACT

The Project "**Temperature and humidity Monitoring System using Arduino Nano and Dht 11 Sensor"** aims to design a simple low-cost, and efficient system to measure and display the surrounding temperature and humidity in real time. The **DHT11 sensor** is used to sense environmental temperature and humidity, which are then processed by the **Arduino Nano** microcontroller. The processed data is displayed as an LCD screen (or serial monitor), providing continuous monitoring of atmospheric conditions. This system is particularly useful in applications such as greenhouse, weather stations, and industrial environments where accurate climate monitoring is essential. The project demonstrates the effective use of embedded system and sensors in creating an affordable environmental monitoring solution.

Furthermore, this project demonstrates the application of embedded system and Internet of Things (IoT) concepts in environmental monitoring. With slight modifications, the system can be enhanced to include data logging, wireless transmissions or integration with cloud platforms for remote monitoring. The successful implementation of this project highlights the importance of microcontroller-based automation in developing smart and sustainable environmental control systems.

Chapter 1

1 INTRODUCTION

Smart Building is the technology of choice that is able to simulate real conditions to be computerized, making it easier to manage or monitor a situation in a building that has a lot of space. Physical computing is a concept to understand the human relationship between environments that are analogous to the digital world. This research aims to design a web-based temperature and humidity monitoring system model using Arduino to monitor room conditions using the internet network. The problem is limited to web design and displays the values of temperature and humidity. The hardware used is Arduino nano, NodeMCU ESP8266 v.3, DHT11 sensor (temperature and humidity). While the software used is Arduino version 1.0.7 for programming the Arduino Uno and javascript, XAMPP microcontrollers which contain Apache, PHP and Mysql

2. MOTIVATION

In the modern era, maintaining accurate environmental conditions is crucial for human comfort, agricultural activities, and various industrial applications. Among the different environmental parameters, temperature and humidity play a vital role in determining the quality of living and working environments. Traditional monitoring methods are often manual, time-consuming, and prone to human error. Hence, there is a strong need for an automatic, reliable, and low-cost monitoring system capable of continuously tracking environmental conditions. This project is motivated by the goal of developing a real-time temperature and humidity monitoring system using Arduino Nano and a DHT11 sensor. The system provides a simple, efficient, and affordable solution to measure and display data accurately. It can also be further enhanced for automation purposes, such as controlling cooling or humidifying devices based on sensor readings

3. LITERATURE REVIEW

Monitoring will provides status information and trends that measurement and repeated evaluations of time to time, monitoring generally done for a specific purpose, for check against the following process object or to evaluate conditions or progress towards management outcome objectives over the effects of actions of some kind among other things, actions for maintain moderate management running Buzzer is an electronic component that functions to convert

DHT11 is a Temperature & Humidity sensor that has a calibrated digital signal output. Using Digital Signal Acqusition technique and temperature and humidity sensing technology, this sensor has high reliability and good long term stability. This sensor includes a resistive humidity measurement component and an NTC temperature measurement component connected to 8-bit microcontroller so that it offers good quality, fast response and anti-interference capabilities

The DHT11 sensor is a sensor that can measure two environmental parameters at once, namely temperature and humidity. In this sensor there is an NTC (Negative Temperature Coefficient) thermistor for measuring temperature, a resistant type humidity sensor and an 8-bit microcontroller that processes the two sensors and sends the results to the output pins in a single-wire bi-directional format (single cable. two-way direction). The sensor has 4 pins, with a PCB breakout that has 3 pins.

Chapter 2

Methodology

The temperature sensor can detect the temperature of the object or the environment and convert these changes into electrical signals that can be interpreted and analyzed.

The humidity sensor mainly measures the humidity of the environment and object and uploads it to the display.

Microcontroller are small, self- contained computer system that can be programmed to control and monitor other devices.

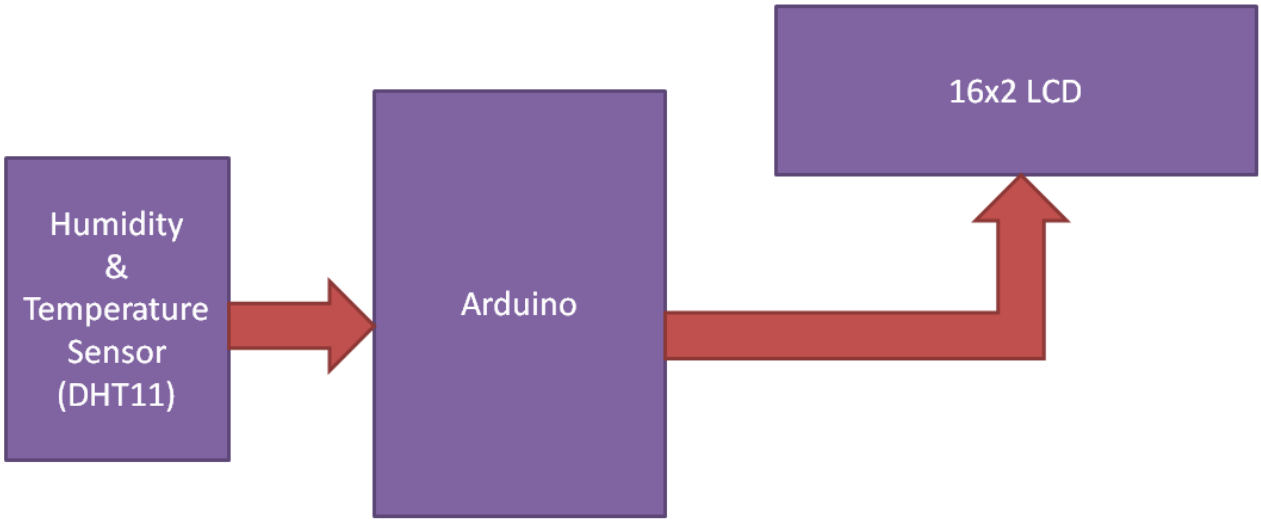
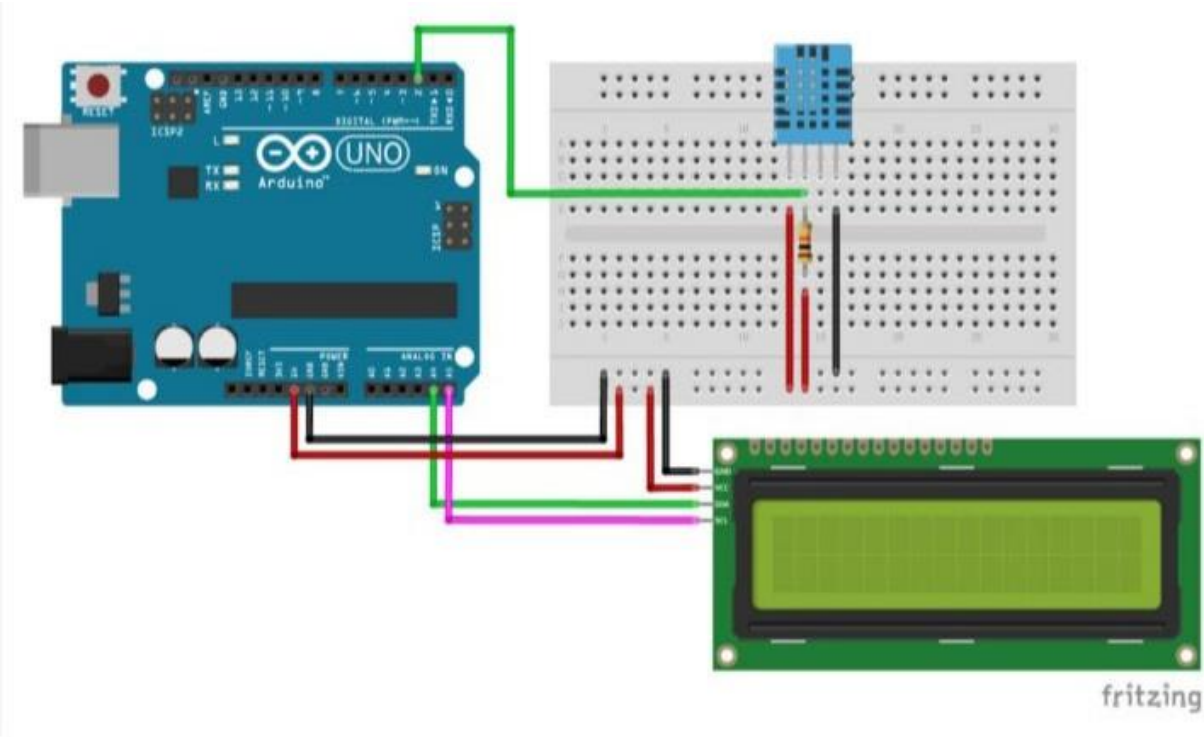


Fig. Block Diagram

Circuit Diagram



Working Principle

The sensor unit measures the temperature and humidity of the considered locations and sends a signal to the controller. The controller process the information obtained and display on LCD display.

Chapter 3

Hardware Components

Sr. No	Component Name	Description	Quantity
1	Ardunio UNO	Ardunio UNO R3 AI mega 328P	1
2	DHT 11 Sensor	DHT 11 Humidity and temperature sensor	1
3	Connecting Wires	Jumper Wires	10
4	Breadboard		1
5	LCD display	16 * 2 LCD	

3.3 Description of Components

1. The Arduino UNO

Figure 3. 1.1 Arduino UNO

Overview:

The Arduino UNO is an open- source microcontroller board based on the Microchip ATmega328P microcontroller (MCU) and developed by Arduino.cc and initially released in 2010. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board has 14 digital I/O pins (six capable of PWM output, 6 analog I/O pins, and its programmable with the Ardunio IDE (Integrated Development Environment), via a type b USB cable. It can be powered by a USB cable or a barrel connector that accepts voltages between 7 and 20 volts, such as a rectangular 9-volts battery. It has the same microcontroller as the Ardunio Nano board, and the same headers as the Leonardo board. The hardware reference design is disturbed under a Creative commons Attributes Share-like 2.5 license and is available on the arduino website. Layout and production files for some versions of the hardware are also available.

2. DHT11 Temperature and Humidity Sensor:

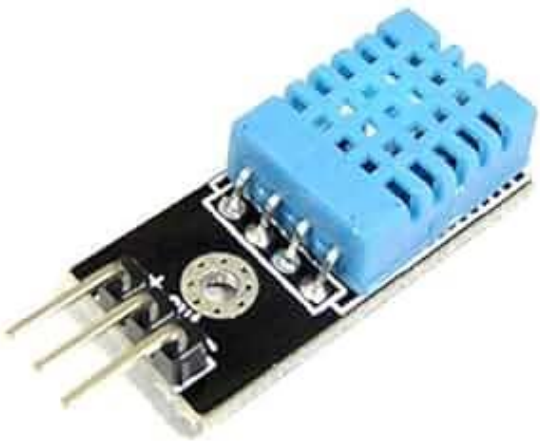


Figure 3.1.2 DHT11 Sensor

The DHT11 is a basic, ultra low-cost digital temperature and humidity sensor. It uses a capacitive humidity sensor and a thermistor, that is a type of resistor whose resistance is strongly dependent on temperature, more so than in standard resistors; to measure the temperature and humidity of the surrounding air and splits out a digital signal on the data pin (no analog input pins needed), which can be displayed on LCD. It is fairly simple to use, but requires careful timing to grab data. The only real downside of this sensors is that we can only get new data from its once 2 second, so when using this sensors, the readings can be up to 2 seconds.

3. LCD Display

Figure 3.1.3 LCD Display

The term LCD stands for liquids crystal display. It is one kind of electronic display module used in an extensive range of applications like various circuits &b devices like mobile phones, calculators, computers, TV sets, etc. These displays are mainly preferred for multi-segment light-emitting diodes and seven segments. The main benefits of using this module are inexpensive; simply programmable, animations and there are no limitations for displaying custom characters, special and even animations, etc.

The LCD pin out is shown below:

- Pin 1 (Ground/Source Pin): This is a GND pin of display, used to connect the GND terminal of the microcontroller unit of power source.
- Pin 2 (VCC/Source Pin): This is the voltage supply pin of the display, used to connect the supply pin of the power source.
- Pin 3 (VO/VEE/Control pin): This pin regulates the differences of the display, used to connect a changeable POT that can supply 0 to 5 v.
- Pin 4 (Register Select/ Control Pin): This pin toggles among command or data register, used to connect a microcontroller unit pin and obtain either 0 or 1 (0 data mode, 1 = command mode).
- Pin 5 (Read/Write/Control pin): This pin toggles the display among the read or write operation, and it is connected to a microcontroller unit pin to get either 0 or 1(0= write operations, and 1 = Read operations)
- Pin 6 (Enable/Control Pins): This pin should be held high to execute Read/Write process, and it is connected to the microcontroller unit & constantly held high.
- Pins 7-14 (Data Pins): These pins are connected into two-wire modes like 4-wires mode and 8-wires mode. In 4 wires mode, only four pins are connected to the microcontroller unit like 0 to 3, whereas inn 8-wires mode, 8-pins are connected to microcontroller unit like 0 to 7.

4. Breadboard

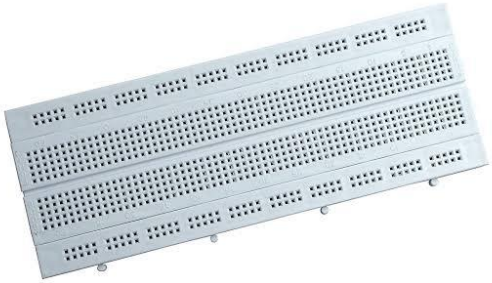


Figure 3.1.4. Bread Board

A breadboard, solderless breadboard, protoboard or terminal array board is a construction base used to build semi-permanent prototypes of electronic circuits. Unlike strip board (Vero board), breadboard do not require soldering to tracks and are hence reusable. For this reason, breadboards are also popular with students and in technological education. A variety of electronic system may be prototyped by using breadboard, from small analog and digital circuits to complete central processing units (CPUs).

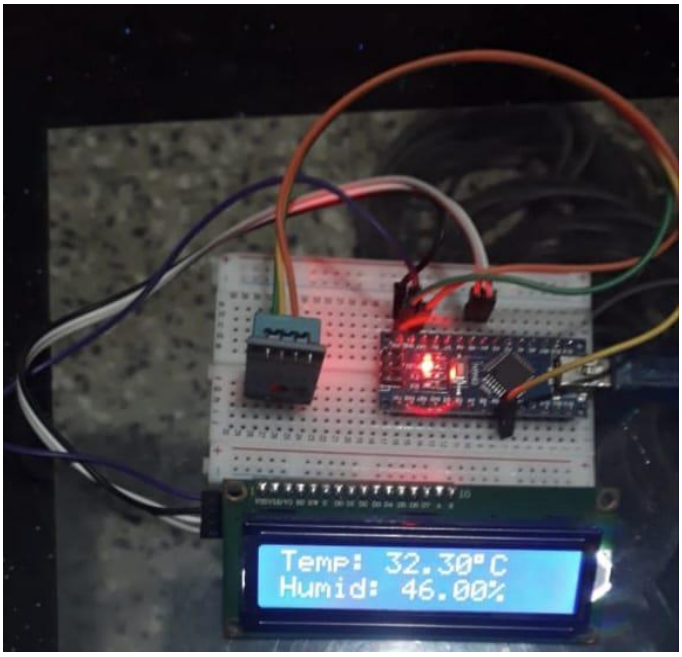
Compared to more permanent circuits connection methods, modern breadboards have high parasitic capacitance, relatively high resistance, and less reliable connections, which are subjects to jostle and physical degradation. Signaling is limited to about 10 MHz, and not everything work properly even well below that frequency.

5.Connecting Wires

Figure 3.1.5 Jumper wires

- Male- male and female-male jumper wires are used.

3.2 Hardware Implementation



The System uses an Arduino Nano as the main controller and a DHT 11 sensor to measure temperature and humidity. The DHT11's data pin is connected to digital pin D2 of the arduino, with VCC to 5 V and GND to GND. The measured values are displayed on a 16 x 2 LCD (via 12C) or on the serial Monitor. The circuit is powered through the Arduino's 5V USB supply, making it compact and efficient for real-time environmental monitoring.

Chapter 4

CODE

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>

// Define I2C LCD address (0x27 or 0x3F depending on your module)
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Define DHT sensor pin and type
#define DHTPIN 7
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

void setup() {
  lcd.init();        // Initialize LCD
  lcd.backlight();   // Turn on LCD backlight
  dht.begin();       // Start DHT sensor

  // Display welcome message
  lcd.setCursor(0, 0);
  lcd.print("Jagdamba College");
  lcd.setCursor(0, 1);
  lcd.print("of Engineering");
  delay(3000);  // Display for 3 seconds

  lcd.clear();
}
```

```
void loop() {
  float h = dht.readHumidity();
  float t = dht.readTemperature(); // Celsius

  // Check for valid readings
  if (isnan(h) || isnan(t)) {
    lcd.setCursor(0, 0);
    lcd.print("Sensor Error!   ");
    delay(1000);
    return;
  }

  // Display temperature
  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print(t);
  lcd.print((char)223); // Degree symbol
  lcd.print("C   ");

  // Display humidity
  lcd.setCursor(0, 1);
  lcd.print("Humid: ");
  lcd.print(h);
  lcd.print("%    ");

  delay(2000); // Update every 2 seconds
}
```

Conclusion

The temperature and humidity Monitoring System using Ardunio Nano and DHT11 successfully measures and display real-time environmental conditions. The project demonstrates an efficient and low-cost solution for monitoring temperature and humidity with good accuracy. It helps in understanding sensor interfacing data acquisition, and microcontroller-based system design. This system can be further enhanced for applications like smart homes, weather stations and industrial monitoring system.

Through this project, the process of interfacing sensors with a microcontroller, collecting digital data, and processing it for meaningful display has been effectively demonstrated. The arduino Nano serves as a compact and powerful controller, while the DHT11 sensor provides stable and accurate readings.

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Kamweru Paul Kuria; Owino Ochieng Robinson; Mutinda Mutava Gabriel — IJERT, Vol. 9, Issue 12, December 2020.

What it does: Uses Arduino Nano + DHT11 + DS3231 (RTC) to log measurements with timestamps; output shown on an LCD.

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What it does: A system measuring temperature & humidity; discusses calibration and accuracy.

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Ammar Yaseen Burjes, Haneafa Yahya Najm, Ghader Salim Aziz — International Journal on Advanced Science, Engineering and Information Technology (IJASEIT), Vol. 11, No. 2, April 2021.

What it does: A portable weather monitoring implementation using Arduino Nano, measuring temperature, humidity (and possibly rainfall), studying thermal behavior.

4. “Temperature and Humidity Control System Model using Arduino Nano”

Yoice R. Putung; Herry S. Langi; Sonny R. Kasenda; Josephin Sundah — International Journal of Computer Applications, Vol. 175, No. 24, 2020.

What it does: Room temperature & humidity control model using Arduino Nano + DHT11; includes web-based monitoring, control with LEDs, a fan.

5. “Prototype Temperature Control Tool to Reduce Humidity on the Inside of the Shoes Based Arduino Nano”

From E-JPTE (Jurnal Elektronik Pendidikan Teknik Elektronika) — Authors designing a wearable prototype that uses Arduino Nano + DHT11 to manage temperature and humidity inside shoes.